



isation (ISRO) in 2019 for Gaganyaan, India's first human spaceflight mission, scheduled for lift-off in 2027. When India decided to send an astronaut to the ISS to gain first-hand experience ahead of Gaganyaan, it was Shukla they selected.

To fly Shukla to the ISS on the Axiom-4 mission, aboard the Crew Dragon capsule launched by a SpaceX Falcon-9 rocket, ISRO spent $500 million. But the experience the astronaut gained in those three weeks is likely to prove invaluable for India's space vision.

That vision includes building the Bharatiya Antariksh Station, India's first orbiting space station, by 2035, followed by plans to land an Indian on the Moon using an indigenous spacecraft by 2040. As Shukla put it, "A lot of things are available on the internet, but when you are part of an activity, you learn things that you don't find in books. I have brought back these experiences with me, and we have already formalised the process of putting them back into the system."

Before Shukla, 285 astronauts of 23 other nationalities had been on the ISS. On this mission, he became the 634th astronaut to orbit Earth. There has also been a recent flurry of near-orbit flights by the likes of billionaires Jeff Bezos and Richard Branson, who have flown to the edge of space in craft their companies have built. But since Yuri Gagarin's pioneering flight in 1961, an average of only 10 astronauts a year have ventured into space.

Shukla is acutely aware of this rare privilege. And like every Indian who has flown into space, he too has come back with a unique experience. Rakesh Sharma had famously said "*Sare jahaan se achha*" when asked how India looked from above. After her first trip to space in 1998, Kalpana Chawla, a US astronaut of Indian origin who tragically died on her second space mission in 2003, had told INDIA TODAY, "More than my Indianness, when I looked up at the stars one night in orbit, I felt I was a resident of the Milky Way."

THE EXPERIENCE SHUKLA GAINED ON THE SPACE EXPEDITION IS ALREADY BEING INTEGRATED INTO THE GAGANYAAN MISSION

Sunita Williams, another US astronaut of Indian origin, carried a Ganesha idol and a copy of the Bhagavad Gita on her first trip to space. "In a little spacecraft zipping around Earth, maybe you are taking for granted where you are and what you are doing," she told this magazine in 2007. "Then, things like Ganesha and the Gita bring you back home and make you feel grounded. It is nice to read about Arjuna's trials and tribulations up there, as it puts your life in perspective."

Shukla, too, has returned with a fount of engrossing reflections. He first saw India when the ISS was making a night pass over the subcontinent. "We were coming from south to north," he recounts in an exclusive interview to INDIA TODAY, "from the Indian Ocean side. Everything was inky black. Then, suddenly, you saw the India peninsula lit up in all its glory. It looked like it was full of sparkling diamonds, with the lights on. It was the most beautiful sight I had ever seen; even now, I get goosebumps just thinking about it."

In an hour-long conversation, Shukla speaks candidly about his rough, risky ride into orbit, the sensation of weightlessness, his space sickness, and how to eat, bathe and use the toilet in near-zero gravity. He recalls the camaraderie he built with his fellow astronauts, the extraordinary thrill of floating around the ISS, the size of which Shukla equates with a 6BHK apartment, conducting scientific experiments and, in his free time, watching through the portals the magnificent sight of Earth below, with multiple sunrises and sunsets in a day, as the ISS circled the planet every 90 minutes.

For Shukla, the overall experience was surreal. "What really hit me was the sheer size of Earth," he says. "You get a very strong sense of your insignificance. I had this peculiar feeling that if everybody on this planet were brought together—all the 8 billion-plus people—we would all probably fit in less than a quarter of Earth. That was a humbling experience." Indeed, his space sojourn has left Shukla with extraordinary insights. He shares them in the exclusive, in-depth interview in the following pages.